

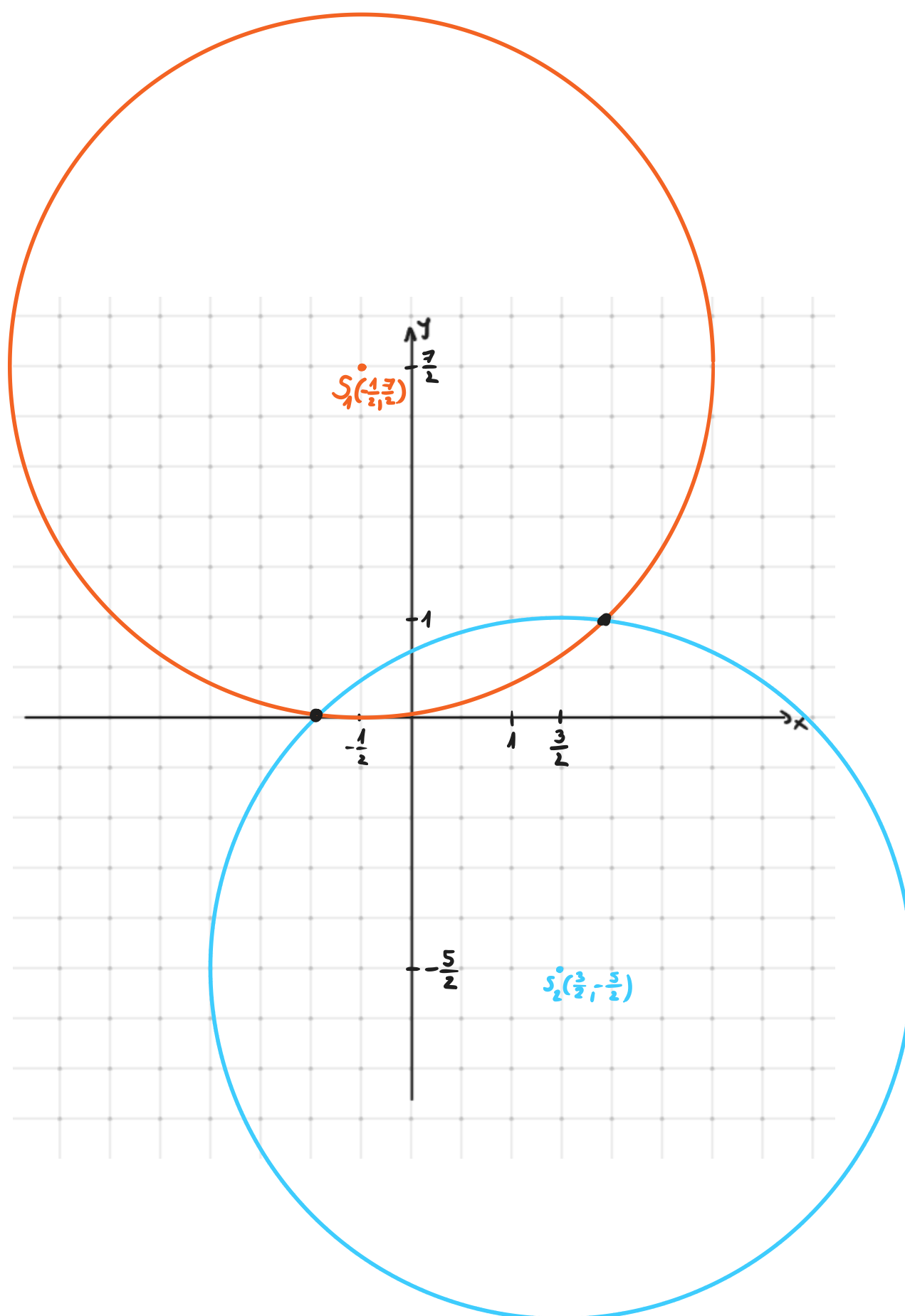
Zadanie z3 (tn) Znajdź współrzędne punktów wspólnych okręgów o równaniach

$$x^2 + y^2 - 3x + 5y - 4 = 0 \text{ i } x^2 + y^2 + x - 7y = 0.$$

Rysunek poglądowy:

$$o_1: x^2 + y^2 - 3x + 5y - 4 = 0 \Rightarrow o_1: (x - \frac{3}{2})^2 + (y + \frac{5}{2})^2 = \frac{25}{2}$$

$$o_2: x^2 + y^2 + x - 7y = 0 \Rightarrow o_2: (x + \frac{1}{2})^2 + (y - \frac{7}{2})^2 = \frac{25}{2}$$



1) Wyznaczamy punkty wspólne z układu równań:

$$\begin{array}{l} \left\{ \begin{array}{l} x^2 + y^2 - 3x + 5y - 4 = 0 \\ x^2 + y^2 + x - 7y = 0 \end{array} \right. \\ \hline -3x - x + 5y - (-7y) - 4 = 0 \\ -4x + 12y - 4 = 0 \Rightarrow x = 3y - 1 \end{array}$$

$$(3y - 1)^2 + y^2 + 3y - 1 - 7y = 0$$

$$9y^2 - 6y + 1 + y^2 + 3y - 1 - 7y = 0$$

$$10y^2 - 10y = 0 \quad |:10$$

$$y(y - 1) = 0$$

$$\begin{array}{l} \left\{ \begin{array}{l} y_1 = 0 \\ x_1 = -1 \end{array} \right. \quad \vee \quad \left\{ \begin{array}{l} y_2 = 1 \\ x_2 = 2 \end{array} \right. \end{array}$$

Odp. $(-1, 0)$ i $(2, 1)$.